

Section A

Answer **all** questions in this section in the spaces provided.

For
Examiner's
use

1 This question is about the chlorides of aluminium, silicon, phosphorus and sulfur.

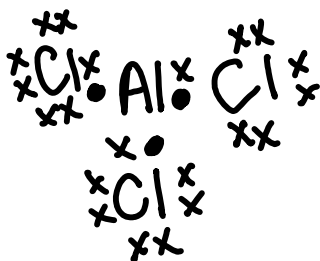
(a) Describe the variations in melting points of the elements aluminium to sulfur and explain the variations in terms of their structures and bonding. [3]

- Aluminium has giant metallic structure held together by strong metallic bonds. During melting, large amounts of energy is required to break the strong metallic bonds, hence aluminium has high melting point.
- Silicon has giant molecular structure held together by numerous strong covalent bonds. As large amount of energy is required to break the strong covalent bonds, hence silicon has high melting point.
- Phosphorus (P_4) and Sulfur (S_8) have simple molecular structure with intramolecular covalent bonds and intermolecular weak van der Waals forces of attractions. In melting, lesser amounts of energy is required to break the weak van der Waals forces of attraction hence P_4 and S_8 have low melting points.

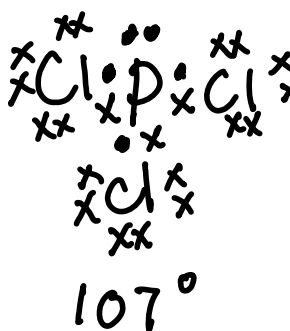
[S_8 has a larger M_r than P_4 and has larger electron cloud which is more easily polarised resulting in stronger and more extensive van der Waals forces of attraction. Thus more energy is required to overcome the intermolecular forces thus, S_8 has a higher melting point than P_4 .]

Aluminium, silicon and phosphorus react with chlorine to form the chlorides of formulae $AlCl_3$, $SiCl_4$, PCl_3 / PCl_5 and S_2Cl_2 respectively.

(b) Draw the dot-and-cross diagrams for $AlCl_3$ and PCl_3 , state the shape and suggest a value for each of the bond angles. [3]



Shape: trigonal planar
Bond angle: 120°

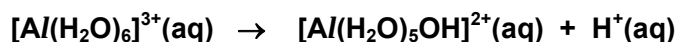


trigonal pyramidal
 107°

- (c) State and explain the pH and the colour observed when some universal indicator is added to the solution formed from AlCl_3 dissolving in water. Write equations where appropriate. [3]

• Solution formed is orange (pH~3-4).

• AlCl_3 undergoes both hydration and hydrolysis as Al^{3+} has a high charge density, hence a high polarising power. It draws electrons away from its surrounding water molecules and weakens the O-H bond.



• 1 mark for both equations

- (d) When liquid SiCl_4 is added to water, a white solid of SiO_2 is immediately formed. Explain why SiCl_4 can be hydrolysed by water to form SiO_2 . [1]

• SiCl_4 is a (simple) covalent chloride that undergoes hydrolysis in water. Hydrolysis occurs due to energetically accessible vacant 3d-orbitals available for dative bonding with water.

- (e) The following table shows three compounds HCN, HCHO and SiCl_4 . From their structures shown, deduce the number of σ bonds around central atom and types of hybridisation in the boxes below. [2]

	HCN	HCHO	SiCl_4
Number of σ bonds around central atom	2	3	4
Type of hybridisation	sp	sp^2	sp^3

- (f) Sulfur and chlorine can be reacted together to form disulfur dichloride, S_2Cl_2 . Disulfur dichloride, S_2Cl_2 , is decomposed by water forming sulfur and a mixture of hydrochloric acid and sulfurous acid.

When 1.35 g of S_2Cl_2 is reacted with an excess of water, 0.48 g of sulfur, S, is produced

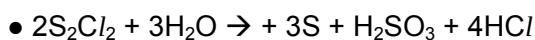
- (i) What is the amount, in moles, of S_2Cl_2 present in 2.70 g?

• Mr of $\text{S}_2\text{Cl}_2 = (2 \times 32.1) + (2 \times 35.5) = 135.2$
 Number of moles $\text{S}_2\text{Cl}_2 = 1.35 / 135.2 = 0.01 \text{ mol}$

- (ii) What is the amount of S produced from 1.0 mol of S_2Cl_2 ?

• 0.01 mol S_2Cl_2 produces $0.48 / 32.1 = 0.0150 \text{ mol}$ of S
 1.0 mol S_2Cl_2 produces $0.0150 \times 1.0 / 0.01 = 1.5 \text{ mol}$ of S

(iii) Construct a balanced equation for the reaction of S_2Cl_2 with water.



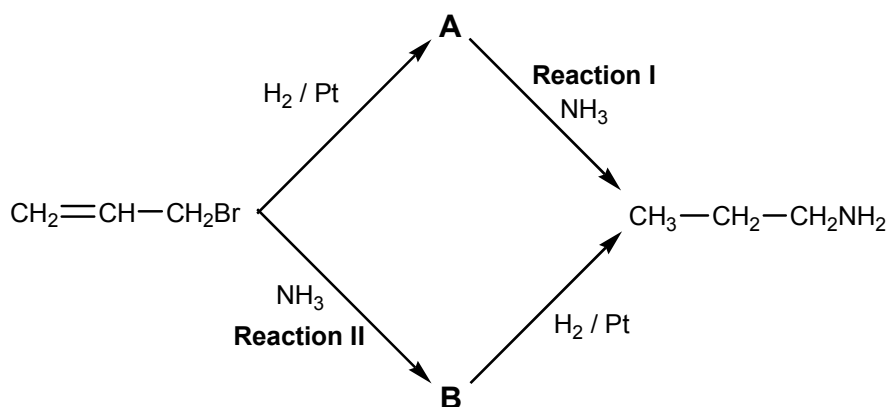
(iv) State and explain the type of reaction in (f)(iii).

- Disproportionation reaction of sulfur as sulfur is oxidised and reduced simultaneously (from +1 in S_2Cl_2 to +6 in H_2SO_3 and 0 in S).

[5]

[Total: 17]

- 2 3-bromopropene is an alkylating agent used in the synthesis of polymers, pharmaceuticals and other organic compounds. 3-bromopropene is a clear liquid with an intense, acrid and persistent smell. 3-bromopropene can be used in the formation of 1-aminopropane through the following two processes.



(a) Give the structural formulae of the compounds **A** and **B**.

[2]

- **A** is $CH_3-CH_2-CH_2Br$
- **B** is $CH_2=CH-CH_2NH_2$

(b) Explain whether 3-bromopropene can show geometric isomerism.

[2]

- For cis-trans isomerism, there must be two different groups bonded to each carbon atom in the $C=C$. Since 3-bromopropene has two H atoms bonded to one of the unsaturated carbon atom in the $C=C$,
- 3-bromopropene **cannot** show geometric isomerism.

- (c) (i) To investigate the kinetics of **Reaction I**, the time taken for a small constant amount of 1-aminopropane was measured when various concentrations of **A** and NH_3 were reacted. The results are shown in the table below:

Experiment	$[\text{A}] / \text{mol dm}^{-3}$	$[\text{NH}_3] / \text{mol dm}^{-3}$	Time for formation of $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ / min
1	0.40	4.00	18
2	0.20	4.00	36
3	0.40	6.00	12

Determine the order of reaction with respect to **A** and NH_3 showing how you arrive at your answers.

Comparing Experiment 1 and 2

- Keeping $[\text{NH}_3]$ constant, when **[A]** is halved, time for formation of $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ is increased by two times => rate of reaction is decreased by two times
Order of reaction with respect to **[A]** is 1

Comparing Experiment 1 and 3

- Keeping **[A]** constant, when $[\text{NH}_3]$ is increased by 1.5 times, time for formation of $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ is decreased by 1.5 times => rate of reaction is increased by 1.5 times.

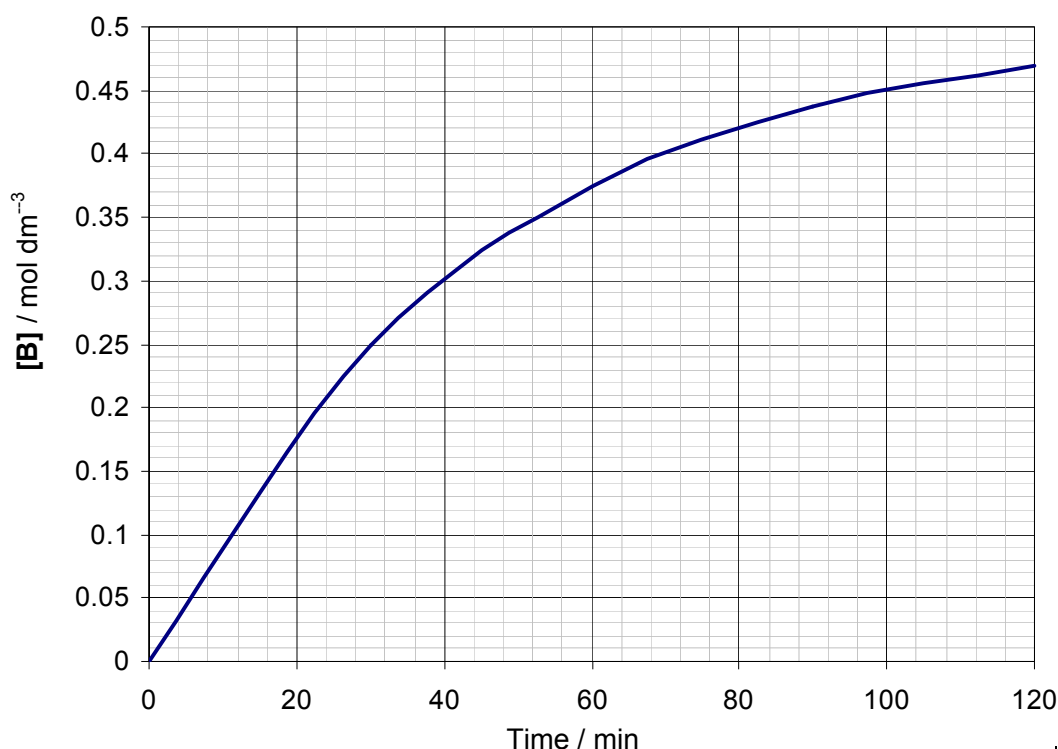
Order of reaction with respect to $[\text{NH}_3]$ is 1

- (ii) Construct the rate equation for the reaction between **A** and NH_3 .

- Rate = $k [\text{A}] [\text{NH}_3]$

[3]

- (d) (i) To investigate the kinetics of **Reaction II**, the concentration of **B** produced with time was obtained plotted on a graph shown below. The initial concentrations of 3-bromopropene and NH_3 were 0.50 mol dm^{-3} and 2.50 mol dm^{-3} respectively. When the reaction is complete, the concentration of B is 0.50 mol dm^{-3} .



When the initial concentration of NH_3 was doubled, the same graph was obtained.

Using the graph and information above, deduce the orders of reaction with respect to NH_3 and 3-bromopropene.

Maximum concentration of **[B]** = 0.50 mol dm^{-3}

- From the graph, the half life of the reaction is constant at 30 min
 - Order of reaction with respect to [3-bromopropene] is 1
- Since the same graph is obtained when $[\text{NH}_3]$ is doubled, it means that $[\text{NH}_3]$ has no effect on the rate of reaction
Order of reaction with respect to $[\text{NH}_3]$ is 0

(ii) Determine the rate constant for the reaction, stating the units.

[4]

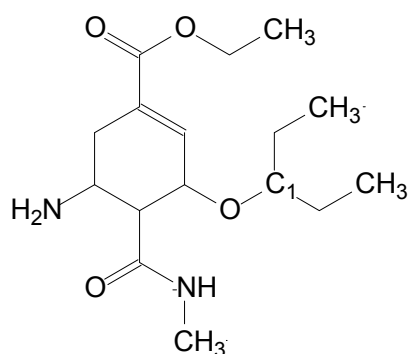
$$t_{\frac{1}{2}} = \frac{\ln 2}{k} \rightarrow k = \frac{\ln 2}{t_{\frac{1}{2}}}$$

$$\bullet k = \frac{\ln 2}{t_{\frac{1}{2}}}$$

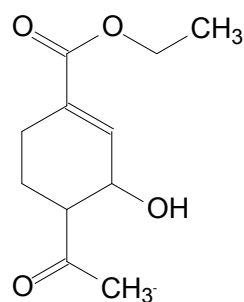
$$= \frac{\ln 2}{30} = 0.0231 \text{ min}^{-1}$$

[Total: 11]

- 3 **Oseltamivir** (Tamiflu) is an antiviral drug that slows the spread of non-resistant strains of the influenza virus between cells in the body. It blocks the action of a viral enzyme called neuraminidase and has since been indicated for the treatment of H5N1 and H1N1 infection. The standard adult dosage is 75 mg twice daily. Compound **C** is a derivative of Oseltamivir that maybe investigated for antiviral activities.



Oseltamivir
M_r = 312.4



Compound C

- (a) (i) A male adult patient has been put on a 1-week oseltamivir treatment. Calculate the total number of moles of oseltamivir taken by this patient over this period of treatment.

$$\bullet \text{ No of moles} = (75 \times 10^{-3}) / 312.4 \times 2 \times 7 = 3.36 \times 10^{-3} \text{ mol}$$

(ii) Describe the shape and state the bond angle about C₁ atom.

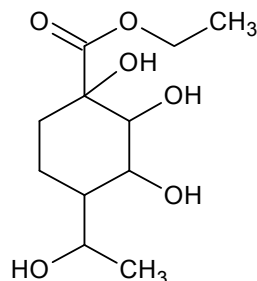
[2]

- Shape: Tetrahedral Bond angle: 109.5°

- (b) (i) Name the functional groups present in Compound **C**.

- Ester, alkene, secondary alcohol, ketone.

- (ii) Compound **D** can be synthesised from **C** using a 2-step reaction. State the reagents and conditions for both reactions. Name the types of reactions involved.



Compound D

Step I:

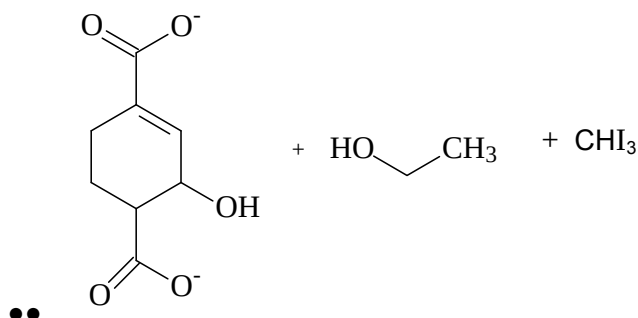
Reagent:	Alkaline MnO_4^-
Condition:	cold
Type of reaction:	oxidation

Step II:

Reagent:	Hydrogen gas
Condition:	Pt catalyst, room temperature
Type of reaction:	reduction

[6]

- (c) (i) Draw the products formed when Compound **C** is reacted with hot alkaline iodine.



- (ii) Describe a chemical test to distinguish **C** from **D**. [4]

- **Test:** Add Br_2/CCl_4 at room temperature to **C** and **D** separately.

Observations:

- **C:** Decolourisation of reddish-brown solution seen.
- **D:** No decolourisation observed.

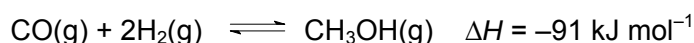
[Total: 12]

Section B

Answer **two** of the three questions in this section on separate paper

- 4 (a) Concern over the ever-increasing use of fossil fuels has led to many suggestions for alternative sources of energy. One of these, suggested by Professor George Olah, winner of a Nobel Prize in chemistry, is to use methanol, CH_3OH , which can be manufactured in a number of ways.

One of the ways to manufacture methanol is from *synthesis gas*, a mixture of CO , CO_2 and H_2 . The CO is reacted with H_2 to form methanol, CH_3OH .



- (i) 0.750 mol each of CO and H_2 is injected into a 2 dm^3 vessel at 400°C and allowed to reach equilibrium. 0.200 mol of CH_3OH was present at equilibrium. Calculate the equilibrium constant, K_c , for the reaction.

	CO(g)	$+ 2\text{H}_2\text{(g)}$	$\rightleftharpoons$	$\text{CH}_3\text{OH(g)}$
Initial/mol	0.750	0.750		0
Change/mol	-0.200	-0.400		+0.200
Equilibrium/mol	0.550	0.350		0.200

$$\begin{aligned}
 K_c &= \frac{[\text{CH}_3\text{OH}]}{[\text{CO}][\text{H}_2]^2} \\
 &= \frac{(0.200 / 2)}{(0.550 / 2)(0.350 / 2)^2} = 11.9 \text{ mol}^{-2} \text{ dm}^6
 \end{aligned}$$

- (ii) The experiment was repeated using a vessel of a different volume and it was found that the equilibrium yield of CH_3OH was higher.

Deduce if the vessel used was larger or smaller.

- Since the equilibrium yield of CH_3OH was higher, the forward reaction which produces lesser number of moles of gaseous particles is favoured. By Le Chatelier's principle, pressure must have been increased / volume has decreased to shift the position of equilibrium to the right.
- Hence, the vessel has a smaller volume.

- (iii) This method of synthesising methanol can be catalysed by a mixture of copper and zirconia oxide. Suggest and explain the effect on the equilibrium composition if another catalyst, MK-121, was used instead.

- There is no effect on the equilibrium composition as catalyst only increases the rate of the forward and backward reaction to the same extent.

[6]

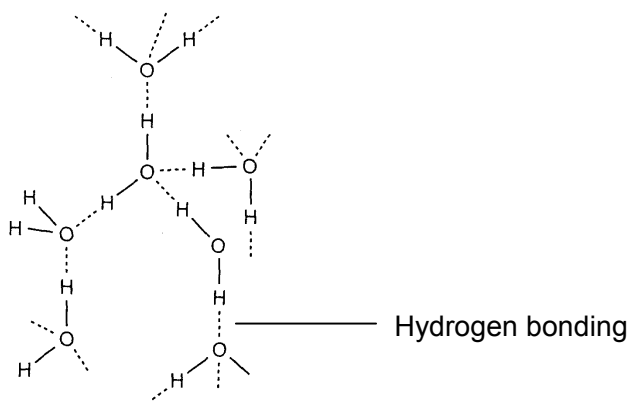
- (b) (i) A mixture of ethane-1,2-diol and water is used as an antifreeze and is found to freeze when temperatures drop below -45°C .

Explain, in terms of structure and bonding, why ethane-1,2-diol is miscible with water.

- Ethane-1,2-diol and water have simple molecular structures. Ethane-1,2-diol has both weak Van der Waals' forces of attraction and hydrogen bonds between its molecules while water molecules has only hydrogen bonds between them.
- Ethane-1,2-diol is miscible with water because the energy released from the formation of hydrogen bonds between ethane-1,2-diol and water is sufficient to overcome the hydrogen bonds in water and also weak Van der Waals' and hydrogen bonding between ethane-1,2-diol molecules.

- (ii) During winter, water freezes when the temperature drops below 0°C and a layer of ice is found at the top of lakes as ice is less dense than water. With the aid of a diagram, explain this phenomenon.

- The presence of two hydrogen atoms and two lone electron pairs in each water molecule allows hydrogen bonding between the water molecules which results in a regular three-dimensional tetrahedral structure in ice. This arrangement in ice creates a very open structure where the molecules are further apart than they are in liquid water. Thus ice is less dense than water.



[4]

- (c) A colour change is observed when methanol is oxidised by hot acidified potassium manganate(VII).

(i) Write a balanced equation for this reaction.



(ii) Explain why methanol has been oxidised.

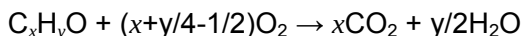
- Methanol undergoes oxidation to form CO_2 where the oxidation number of C changes from -2 in methanol to +4 in CO_2 .

[2]

- (d) 0.00754 mol of compound **E**, $\text{C}_x\text{H}_y\text{O}$, was burnt in excess oxygen to produce 1.66 g of carbon dioxide and 0.54 g of water.

E decolourises aqueous bromine and reacts with PCl_5 giving off white fumes. Upon refluxing **E** with acidified potassium manganate(VII), a symmetrical product **F**, $\text{C}_5\text{H}_6\text{O}_5$, is formed. **F** does not give a red precipitate with Fehling's solution but a orange precipitate is observed with 2,4-dinitrophenylhydrazine. 1 mol of **F** also reacts with 1 mol of Na_2CO_3 with effervescence observed. **F** reacts with NaBH_4 to form **G**.

(i) Using the data above, verify that the molecular formula of **E** is $\text{C}_5\text{H}_8\text{O}$.

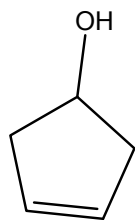


- No of moles of $\text{CO}_2 = 1.66/44 = 0.0377$ mol
No of moles of $\text{H}_2\text{O} = 0.54/18 = 0.0300$ mol
- By comparing mole ratio, $\frac{1}{x} = \frac{0.00754}{1.66/44}$ $\frac{2}{y} = \frac{0.00754}{0.54/18}$

Hence, $x = 5$ and $y = 8$

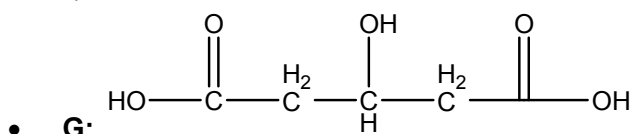
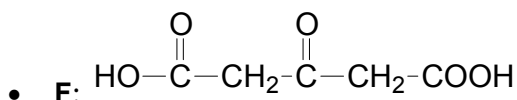
(ii) Deduce the structures of compounds **E**, **F** and **G**, explaining the chemistry of the reactions involved.

- E** decolourises aqueous bromine $\rightarrow$ $\text{C}=\text{C}$ present
- E** reacts with PCl_5 giving off white fumes of HCl $\rightarrow$ $-\text{OH}$ present
- F** does not give a red precipitate with Fehling's solution but form a orange precipitate with 2,4-dinitrophenylhydrazine $\rightarrow$ **F** is a ketone
- E** is a 2° alcohol
- 1 mol of **F** also reacts with 1 mol of $\text{Na}_2\text{CO}_3 \rightarrow 2$ $-\text{COOH}$ groups present
- F** reacts with NaBH_4 to form **G** $\rightarrow$ carbonyl group in **F** is reduced but not the carboxylic acid groups.



• **E:**

(**E** undergoes oxidation with KMnO_4 where 2° alcohol is converted to ketone and $\text{C}=\text{C}$ in a ring undergoes total bond cleavage to form **F**)

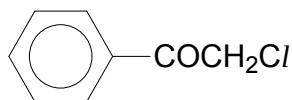


(9 marking points: Maximum 6)

[8]

[Total: 20]

- 6 (a) Chloroacetophenone was formerly the most widely used tear gas, under the codename *CN*. It was used in warfare and in riot control.



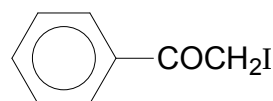
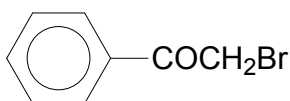
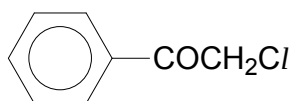
The efficiency of a tear gas is expressed by its 'intolerable concentration'. The intolerable concentration of the tear gas *CN* has been measured as 0.030 g m^{-3} of air.

How many moles of chloroacetophenone need to be sprayed into a room of volume 60 m^3 in order to achieve this concentration?

[2]

- Mass of *CN* needed = $0.03 \times 60 = 1.8 \text{ g}$
 M_r of *CN* = 154.5
- No of moles of chloroacetophenone needed = $1.8/154.5 = 0.0117 \text{ mol}$

- (b) The compounds below differ in their physical properties and reactivities.



(i) Suggest and explain how the boiling points of these three compounds differ.

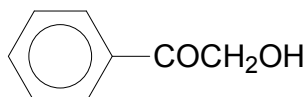
- The boiling points increase from left to right.
- The compounds are all covalent with simple molecular structures. Boiling involves breaking weak van der Waals' forces of attraction between molecules. The M_r of the compounds increases from left to right leading to an increase in the electron cloud. This causes an increase in the polarization of the electron cloud resulting in stronger van der Waals' forces. Hence more energy is required to overcome the van der Waals' forces leading to a higher boiling point.

(ii) Suggest and explain how the C-X bond polarities in these three compounds differ.

- The C-Cl bond is the most polar and C-I bond is the least polar.
- Cl is the most electronegative and I the least electronegative. As a result, the bond pair of electrons in the C-Cl bond is drawn closer to Cl, resulting in a greater charge separation and a more polar bond.

(iii) Describe and explain how the reactivities of the three compounds differ towards reaction with aqueous sodium hydroxide.

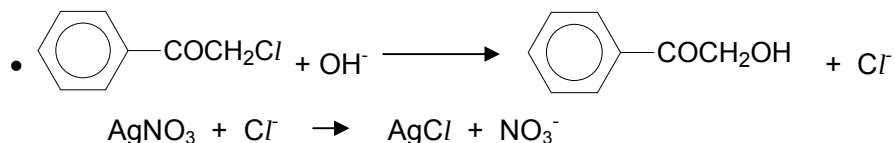
- The substitution reaction requires the breaking of the C-X bond. The 3 compounds undergo substitution reaction with aqueous sodium hydroxide to form



- C-X bond length increases from Cl to Br to I due to increasing size of the halogen atom. The shorter the bond length, the stronger the bond and more energy is required to break the bond resulting in low reactivity with aq NaOH.
- Thus reactivity increases from Cl to I as the C-X bond length increases.

(iv) Describe how you would distinguish the three compounds above, writing balanced equations for reactions mentioned.

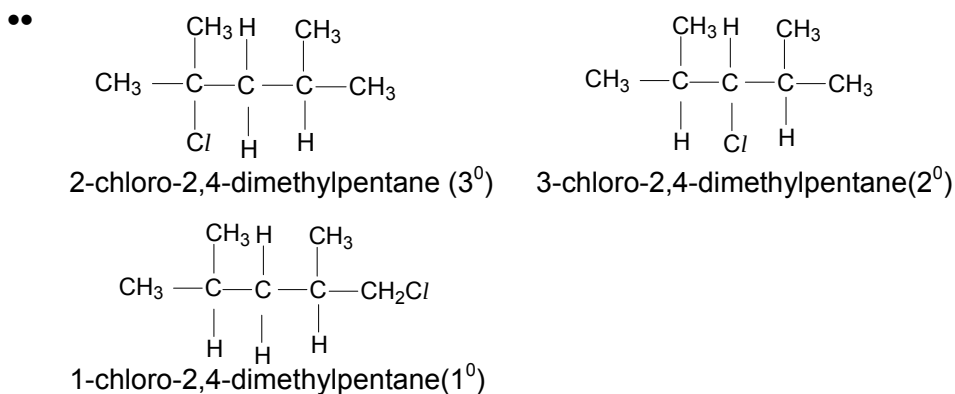
- Boil all three with aq. NaOH and acidify with dilute nitric acid. Add aq. AgNO_3 to all three mixtures.
- A white ppt will be observed if Cl is present, a pale yellow ppt if Br is present and a yellow ppt if I is present.



[10]

- (b) Upon irradiation of ultra-violet light, 2,4-dimethylpentane undergoes monochloro-substitution.

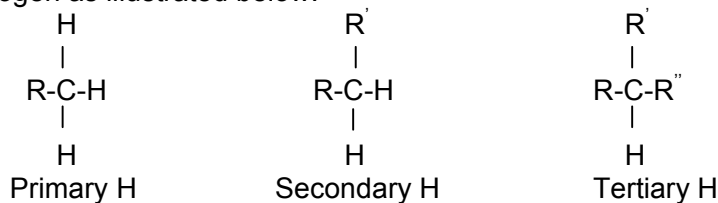
- (i) Draw the structural formula of all the possible monochlorinated products.



- (ii) Predict the mole ratio of the monochlorinated products in (c)(i).

• mole ratio of 1⁰, 2⁰ and 3⁰ monochlorinated products 6 : 1 : 1

- (iii) The hydrogen in an organic molecule can be classified as primary, secondary or tertiary hydrogen as illustrated below.



You are provided with the following bond dissociation enthalpy data:

Bond	Bond dissociation enthalpy / kJ mol ⁻¹
H-Cl	+432
Cl-Cl	+243
H-CH ₂ CH ₃	+420
H-CH(CH ₃) ₂	+401
H-C(CH ₃) ₃	+390
Cl-CH ₂ CH ₃	+338
Cl-CH(CH ₃) ₂	+339
Cl-C(CH ₃) ₃	+330

Using the equation for the monochlorination of an alkane below, estimate the respective enthalpy changes of the monochlorination of primary, secondary and tertiary hydrogen atoms in 2,4-dimethylpentane.



$$\Delta H = -\Delta H_{\text{RCl}} - \Delta H_{\text{HCl}} + \Delta H_{\text{RH}} + \Delta H_{\text{Cl-Cl}}$$

• For 1° Hydrogen $\Delta H = -107 \text{ kJ mol}^{-1}$

• For 2° Hydrogen $\Delta H = -127 \text{ kJ mol}^{-1}$

• For 3° Hydrogen $\Delta H = -129 \text{ kJ mol}^{-1}$

- (iv) From your answer in (c)(iii), suggest a factor that could have affected the yield of the monochlorinated products from 2,4-dimethylpentane. Explain your answer.

• From (c)(iii), the enthalpy change becomes more negative for the substitution of 1° to 3° hydrogen implying that the formation of 2-chloro-2,4-dimethylpentane is more spontaneous.

• A factor that could give rise to this result is the strength of C-H bonds.

[8]

[Total: 20]

- 7 (a) Sodium hydroxide is often used for acid base titrations. A student prepared a standard solution of sodium hydroxide by dissolving 10.0 g of sodium hydroxide in 250 cm³ of water. During the preparation, he found the solution warm to the touch.

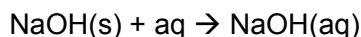
- (i) Calculate the concentration of the sodium hydroxide solution prepared and calculate its pH.

$$\text{Number of moles of NaOH} = \frac{10.0}{40} = 0.25 \text{ mol}$$

• Concentration of NaOH = $0.25/250 \times 1000 = 1 \text{ mol dm}^{-3}$
 $\text{pOH} = -\lg[\text{OH}^-] = 1$

• $\text{pH} = 14 - 1 = 13$

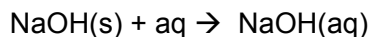
- (ii) Given the information below, calculate the heat evolved when solid sodium hydroxide is dissolved in water.



	$\Delta H_f / \text{kJ mol}^{-1}$
NaOH (s)	-425
NaOH (aq)	-470

Hence, find the increase in temperature of the standard solution.

[6]



• $\Delta H = \Delta H_{\text{f of NaOH (aq)}} - \Delta H_{\text{f of NaOH (s)}} = -470 - (-425) = -45 \text{ kJ mol}^{-1}$

$$\text{Number of moles of NaOH} = \frac{10.0}{40} = 0.25 \text{ mol}$$

$$\bullet \text{ Heat evolved} = 45 \times 0.25 = 11.25 \text{ kJ}$$

$$\bullet \text{ Temperature increase} = \frac{Q}{mc} = \frac{11.25 \times 1000}{(250)(4.18)} = 10.8^\circ \text{C}$$

(b) Carboxylic acids are *weak Bronsted acids*.

(i) Define what you understand by the term "*weak Bronsted acids*".

- *Bronsted acids* are proton donors. Weak acids dissociates partially to its ions.

The following table gives the pK_a values of three organic compounds.

Compound	M_r	pK_a	Melting point / $^\circ\text{C}$
Propan-1-ol	60.0	16.0	-126
Ethanoic acid	60.0	4.79	17
Chloroethanoic acid	94.5	2.86	63

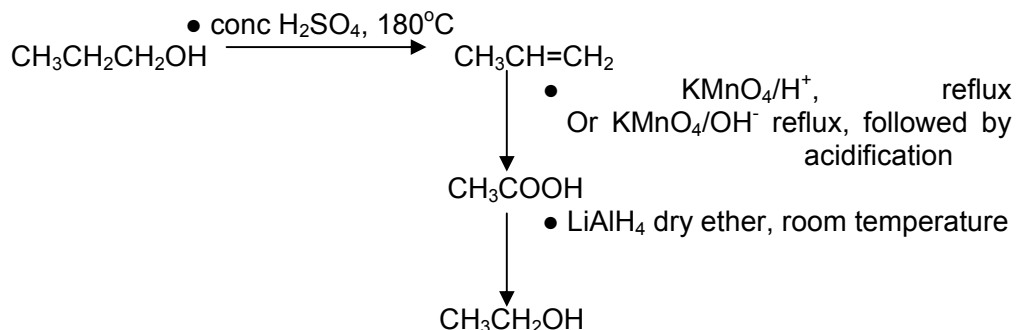
(ii) Arrange the organic compounds in increasing acidity. Explain your answer.

- In increasing acidity: Propan-1-ol < Ethanoic acid < Chloroethanoic acid
- Ethanoic acid and chloroethanoic acid are more acidic as the carboxylate anion (RCOO^-) is more stable than the alkoxide ion (RO^-) due to the delocalisation of negative charge over two electronegative oxygen atoms, resulting in a much more stabilised ion as the negative charge is better dispersed and is more likely to release H^+ .
- RO^- has its negative charge localized on the oxygen atom hence is less likely to release H^+ thus least acidic
- Chloroethanoic acid is more acidic than ethanoic acid as the electron withdrawing Cl further disperses the negative charge on the carboxylate anion (RCOO^-). $\text{ClCH}_2\text{COO}^-$ is more stabilised as the negative charge is better dispersed and is more likely to release H^+ .

(iii) Explain, in terms of bonding and structure, the difference in melting point between propan-1-ol and ethanoic acid.

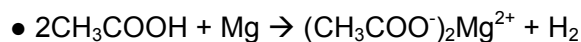
- Both propan-1-ol and ethanoic acid have simple molecular structure held together by hydrogen bonds and weak van der Waals forces of attraction between the molecules.
- Ethanoic acid has a higher melting point as it has more extensive hydrogen bonding compared to propan-1-ol. Ethanoic acid is also able to form dimers thus having a larger M_r and have stronger and more extensive van der Waals forces of attraction.

- (iv) Suggest a 3-step synthesis of ethanol from propan-1-ol.



[1m for each reagent & condition & correct intermediate]

- (v) Write a balanced equation for the reaction between ethanoic acid and magnesium metal. [11]



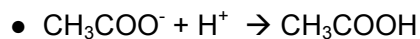
- (c) Mixture J is a *buffer* that is made up of 50 cm³ of 1 mol dm⁻³ sodium hydroxide solution and 100 cm³ of 1 mol dm⁻³ of ethanoic acid.

- (i) What is meant by a *buffer*?

• Buffers are solutions that resist changes in pH upon additions of small amounts of acid or alkali.

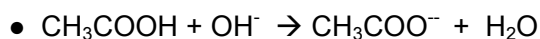
- (ii) Write equations to show how mixture J acts as a buffer.

When small amounts of acid is present,



H⁺ is removed by CH₃COO⁻ mainly from the salt thus there is negligible change in pH.

When small amounts of base is present,



OH⁻ is removed thus there is negligible change in pH.

[3]
[Total: 20]